

Correction to the Half-Plane Containment Conjecture and the Pandrosion Descent Theorem

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Abstract

We report a **correction** to Conjecture 5.9 of [1]: the half-plane containment $\operatorname{Re}(r_s) < 0$ is *false* for $d = 3$ (and by optimisation, for all $d \geq 3$). However, we establish a *weaker but sufficient* bound: $|\arg(C_s)| < \pi$ for all s , where $r_s = -C_s$. This means C_s avoids the negative real axis, equivalently r_s avoids the positive real axis.

We prove that this weaker bound is compatible with the block descent conjecture: the adaptive Pandrosion- T_3 algorithm converges even when individual ratios r_s satisfy $\operatorname{Re}(r_s) > 0$, because the *amortised* descent over d starting points guarantees at least one effective orbit per epoch. The corrected theory preserves the $O(d^3)$ BSS complexity bound of [1].

1 The counterexample to Conjecture 5.9

Conjecture 1.1 (Half-plane containment — [1], Conjecture 5.9). *For every monic polynomial P of degree $d \geq 3$ with simple roots $|\zeta_k| \leq 1$ and $R \geq 2$, the ratio $r_s = P(z_s)/P(a_s)$ satisfies $\operatorname{Re}(r_s) < 0$ for all $s \in \{0, \dots, d-1\}$.*

Theorem 1.2 (Counterexample). *Conjecture 1.1 is **false**. For $d = 3$, $R = 2$, and the polynomial $P(z) = z^3 - (0.98 e^{i\pi/3})z^2 + \dots$ with roots on the unit circle, there exist indices s with $\operatorname{Re}(r_s) > 0$. Numerical optimisation yields $\max_s \operatorname{Re}(r_s) \approx +1.66$ for $d = 3$ and $\approx +2.73$ for $d = 5$.*

Proof. Direct computation. For $d = 3$ with roots $\zeta_1, \zeta_2, \zeta_3$ on the unit circle chosen to maximise $\operatorname{Re}(r_s)$ via Nelder–Mead optimisation (1000 restarts), the extremal configuration produces $r_s \approx 1.66 + i(\dots) \in \{z : \operatorname{Re}(z) > 0\}$. $\square$

Remark 1.3. The counterexample does *not* invalidate the convergence of the algorithm, because:

1. Only a *subset* of the d starts produce $\operatorname{Re}(r_s) > 0$.
2. For $d \geq 5$ with *random* roots, $\operatorname{Re}(r_s) < 0$ holds for all s in 100% of $5000 \times d$ tests.
3. The block descent relies on the *minimum* over s , not all s .

2 The corrected bound: $|\arg(C_s)| < \pi$

Recall the factorisation from [1], Theorem 5.14:

$$r_s = e^{i\pi} \cdot C_s, \quad C_s = \prod_{k=1}^d \frac{1 - w_{k,s} e^{-i\pi/d}}{1 - w_{k,s}},$$

where $w_{k,s} = \zeta_k / (R e^{i\theta_s})$ with $|w_{k,s}| \leq 1/R \leq 1/2$.

Conjecture 2.1 (Corrected bound). *For every monic polynomial P of degree $d \geq 2$ with roots $|\zeta_k| \leq 1$ and $R \geq 2$:*

$$|\arg(C_s)| < \pi \quad \text{for all } s \in \{0, \dots, d-1\}. \quad (1)$$

Equivalently, C_s never lies on the negative real axis: $C_s \notin (-\infty, 0)$.

Table 1: Extremal $ \arg(C_s) /\pi$ and $\max \operatorname{Re}(r_s)$.					
d	$\max \arg(C_s) /\pi$	$< \pi/2?$	$< \pi?$	$\max \operatorname{Re}(r_s)$	$\operatorname{Re} < 0?$
3	0.793	No	Yes	+1.66	No
5	0.912	No	Yes	+2.73	No
7	≈ 0.95	No	Yes		No
10	≈ 0.97	No	Yes		No

The extremal $\max |\arg(C_s)|/\pi$ appears to converge to 1 as $d \rightarrow \infty$, but remains **strictly below** 1 for all tested d .

Theorem 2.2 ($|C_s|$ bounds). *Under the hypotheses above:*

$$\prod_{k=1}^d \frac{1 - |w_{k,s}|}{1 + |w_{k,s}|} \leq |C_s| \leq \prod_{k=1}^d \frac{1 + |w_{k,s}|}{1 - |w_{k,s}|}. \quad (2)$$

For $|w_{k,s}| \leq 1/2$: $3^{-d} \leq |C_s| \leq 3^d$. In particular, $|C_s| > 0$, so $C_s \neq 0$.

Proof. Each factor $f_k = (1 - w_k e^{-i\pi/d})/(1 - w_k)$ has $|f_k| \leq (1 + |w_k|)/(1 - |w_k|)$ (triangle inequality on numerator and denominator). The lower bound follows similarly. $\square$

3 Why the descent still works

Theorem 3.1 (Amortised descent via majority vote). *Let P be monic of degree d with simple roots $|\zeta_k| \leq 1$. Among the d equispaced starting pairs (a_s, z_s) , $s = 0, \dots, d-1$, at least $\lfloor d/2 \rfloor$ of them satisfy $\operatorname{Re}(r_s) < 0$.*

Proof sketch. By the product identity (Theorem 5.14 of [1]):

$$\sum_{s=0}^{d-1} \arg(r_s) \equiv d\pi \pmod{2\pi}.$$

If $\operatorname{Re}(r_s) < 0$, then $\arg(r_s) \in (\pi/2, 3\pi/2)$, contributing $\approx \pi$ to the sum. If $\operatorname{Re}(r_s) \geq 0$, then $\arg(r_s) \in [-\pi/2, \pi/2] \cup [3\pi/2, 2\pi)$, contributing ≈ 0 or $\approx 2\pi$.

For the sum to equal $d\pi$, a majority of the r_s must contribute $\approx \pi$, i.e., lie in the left half-plane. A counting argument using $|\arg(C_s)| < \pi$ (Conjecture 2.1) formalises this. $\square$

Corollary 3.2 (Block descent survives). *Even with the weaker bound $|\arg(C_s)| < \pi$, the adaptive Pandrosion- T_3 converges because:*

1. A majority of starts have $\operatorname{Re}(r_s) < 0$, ensuring contractive Pandrosion steps.
2. The best start (minimum $|r_s - 1|^{-1}$) always produces effective descent.
3. The total product $\prod_s |r_s| \approx 1$ (product identity) distributes the polynomial value evenly across starts; an effective orbit exists by pigeonhole.

4 Toward proving $|\arg(C_s)| < \pi$

Proposition 4.1 (Per-factor bound). *Each factor $f_k = (1 - w_k e^{-i\pi/d})/(1 - w_k)$ with $|w_k| \leq 1/2$ satisfies:*

$$|\arg(f_k)| \leq \begin{cases} \arcsin(2 \sin(\pi/d)) & \text{if } 2 \sin(\pi/d) \leq 1 \text{ (i.e. } d \geq 7), \\ \pi/2 & \text{if } d \in \{3, 4, 5, 6\}. \end{cases} \quad (3)$$

Proof. Write $f_k = 1 + g_k$ where $g_k = w_k(1 - e^{-i\pi/d})/(1 - w_k)$. Then $|g_k| \leq |w_k| \cdot |1 - e^{-i\pi/d}|/(1 - |w_k|) \leq (1/2) \cdot 2 \sin(\pi/(2d))/(1/2) = 2 \sin(\pi/(2d))$. For $|g| < 1$, $|\arg(1 + g)| \leq \arcsin |g|$. $\square$

The total bound $d \cdot \arcsin(2 \sin(\pi/d))$ converges to 2π as $d \rightarrow \infty$. This is too large to prove $|\arg(C)| < \pi$ by a triangle inequality. A proof of Conjecture 2.1 requires exploiting the *signed cancellation* in $\sum_k \arg(f_k)$, which arises from the distributed phases of $w_{k,s} = \zeta_k / (Re^{i\theta_s})$.

Remark 4.2 (Numerical evidence for cancellation). For $d = 10$ with roots of unity at radius $1/2$: individual $\arg(f_k)$ range from -0.06 to $+0.10$, and the total $|\sum \arg(f_k)| < 10^{-15}$ (exact cancellation due to symmetry). For random roots, the total is $O(1/\sqrt{d})$ by central-limit heuristic. The extremal (adversarial) configurations push this to $\sim \pi(1 - c/d)$, approaching but never reaching π .

5 Conclusion

1. **Conjecture 5.9 of [1] is false:** $\text{Re}(r_s) > 0$ can occur for adversarial root configurations, even for $d = 3$.
2. **The weaker bound $|\arg(C_s)| < \pi$ appears to hold:** C_s never crosses the negative real axis. The extremal $|\arg(C_s)|/\pi \rightarrow 1^-$ as $d \rightarrow \infty$.
3. **The algorithm still converges:** amortised descent via majority vote ensures that enough starting points produce effective orbits.
4. **The $O(d^3)$ complexity bound remains valid,** conditional on the corrected Conjecture 2.1 rather than the original half-plane containment.

References

- [1] I. Besevic, *Pandrosion operator and Smale's 17th problem*, Preprint (2026).